Fundamentals of Circuits and Electronics
CSAI 205

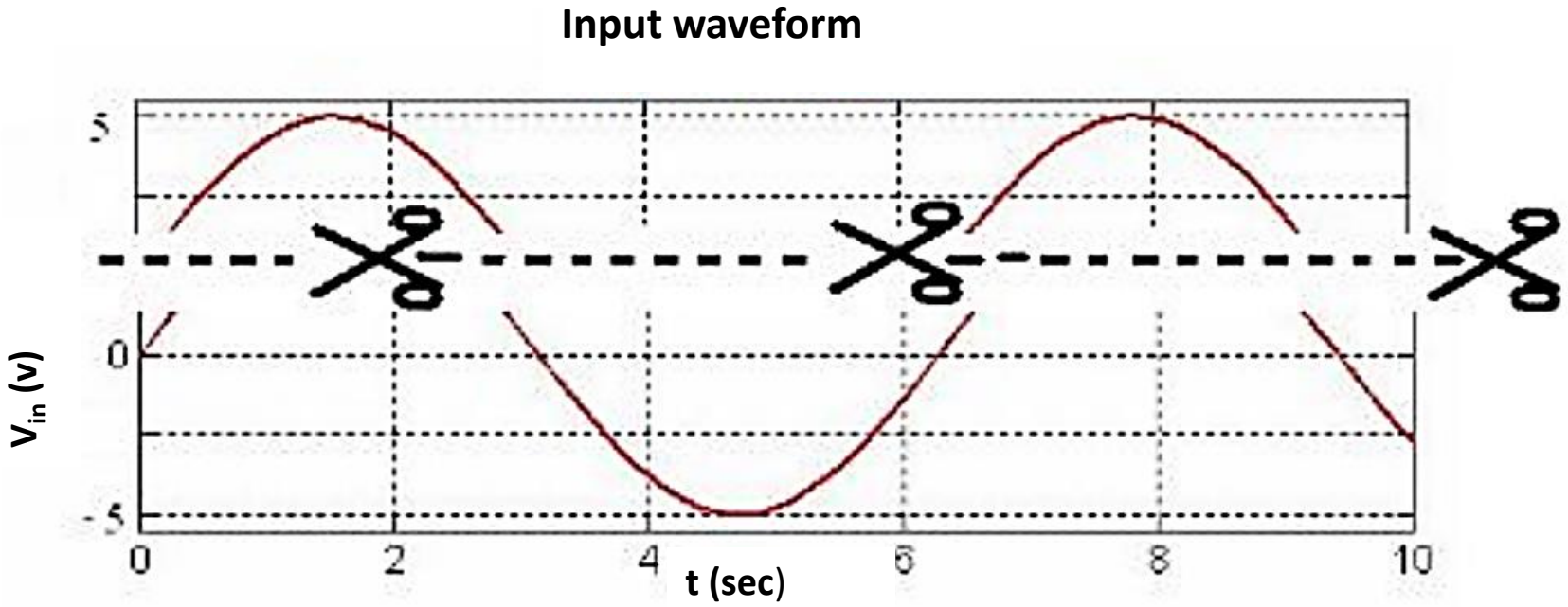
Assoc. Prof. **Mohamed Maher Ata**
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momaher@zewailcity.edu.eg

Diode Applications: Continue

1. **Rectification→ Previous lecture**
2. **Clippers**
3. **Clampers**
4. **Voltage multipliers**

Today's lecture

(2) Clippers

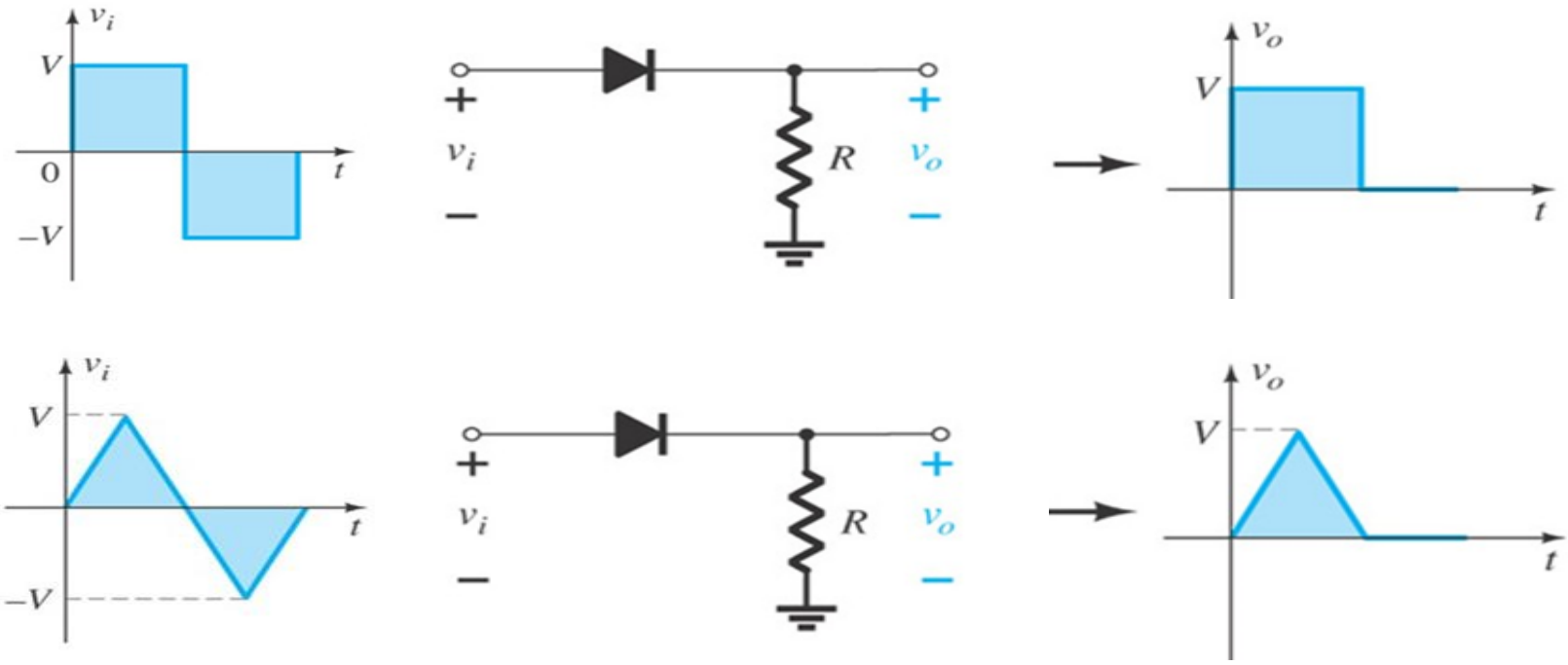


Real time applications

1. **Signal Conditioning in Communication Systems**
   - **Noise Reduction:** Clippers help in removing noise spikes from transmitted signals in radio and telecommunication systems.
   - **Amplitude Limiting:** They ensure that signals remain within a defined range to prevent overloading of circuits.
2. **Wave Shaping in Electronics**
   - **Square Wave Generation:** Clippers modify sine waves into square waves for use in digital electronics and waveform generators.
   - **Pulse Shaping Circuits:** Used in digital circuits to create sharp pulse waveforms.
3. **Voltage Protection in Power Supplies**
   - **Overvoltage Protection:** Clippers prevent excessive voltage from damaging sensitive electronic components.
   - **Surge Suppression:** They act as protection circuits against voltage spikes in power lines.
4. **Biomedical Applications**
   - **ECG and EEG Signal Processing:** Clippers are used in medical devices to remove unwanted high-amplitude signals, ensuring accurate readings of heartbeats and brain activity.
5. **Audio Signal Processing**
   - **Peak Clipping in Amplifiers:** Used in audio processing to prevent distortion by limiting signal peaks.
   - **Noise Filtering:** Helps in removing unwanted high-voltage spikes in audio signals.
6. **Radar and RF Systems**
   - **Amplitude Control:** Clippers help regulate signal levels in radar and RF circuits to prevent saturation of receiver components

Diode Clippers

Half wave rectifier acts as clipper



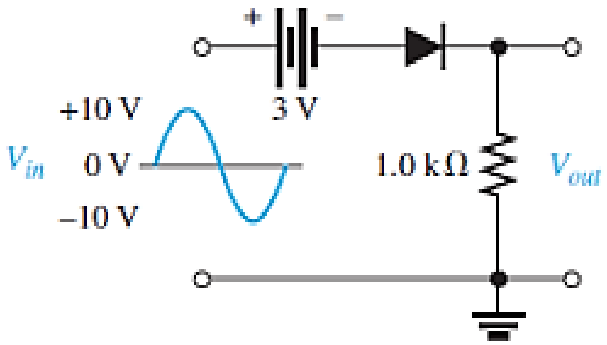
Biased Clippers

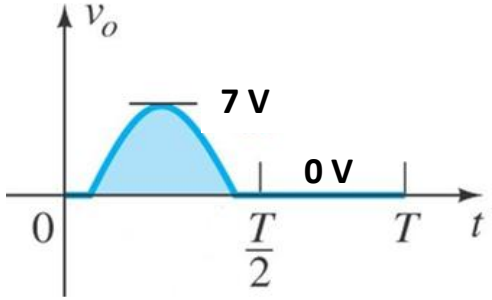
Adding a DC source in series with the clipping diode changes the effective forward bias of the diode.

Example: Determine the output waveform for the network

(Assume **ideal** diode)

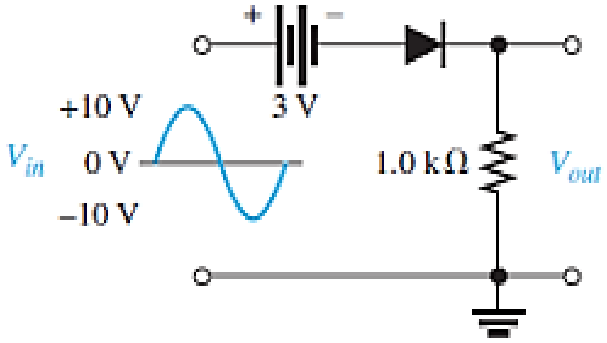
+ve half: $+10 - 3 - Vo = 0$
−ve half: $zener\ is\ open, I = 0, V = 0$

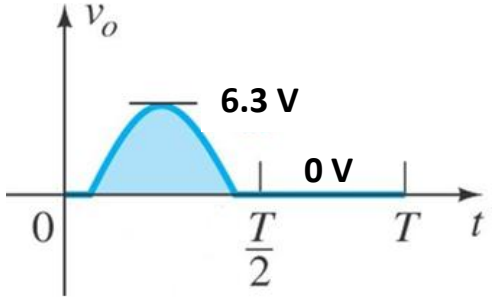




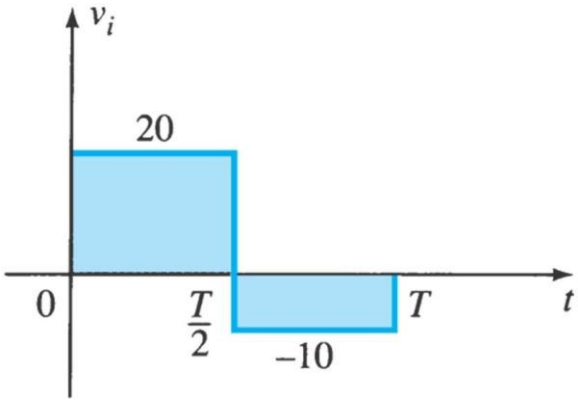
(Assume **practical** diode)

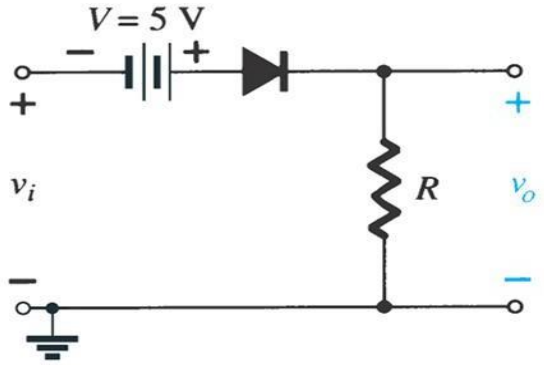
+ve half: $+10 - 3 - 0.7 - Vo = 0$
−ve half: $zener\ is\ open, I = 0, V = 0$

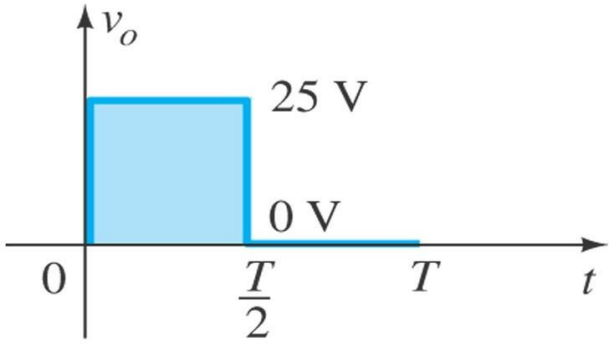


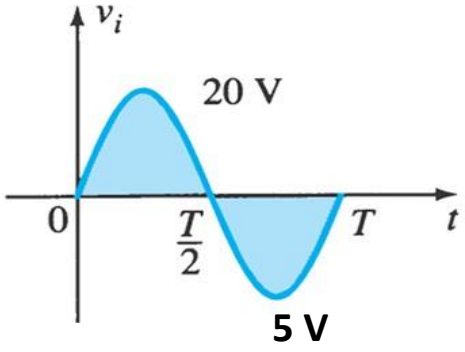


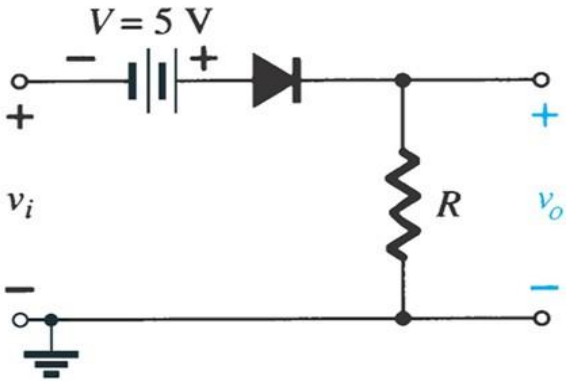
Example: Determine the output waveform for the network (assume ideal diode).

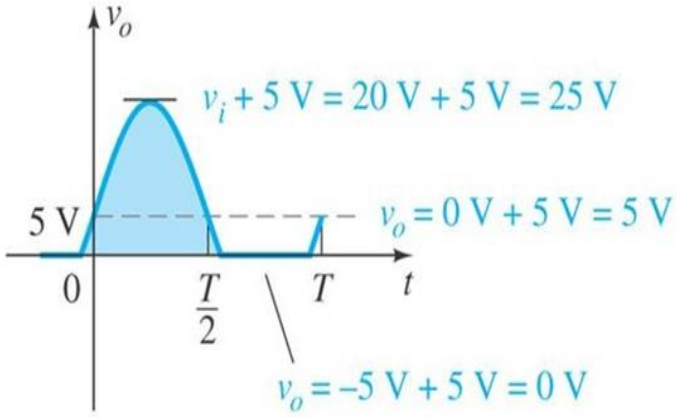




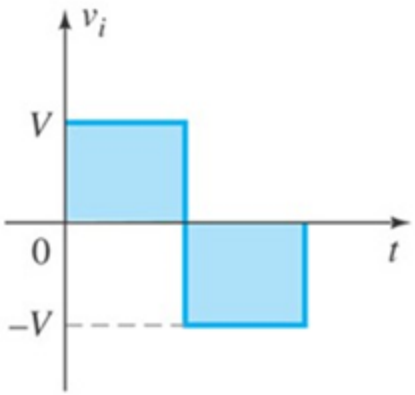


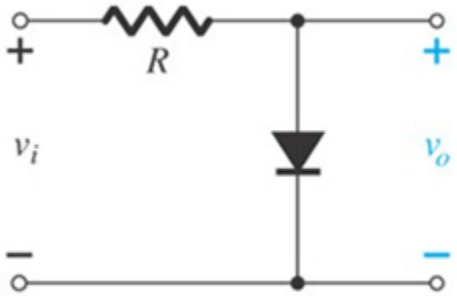


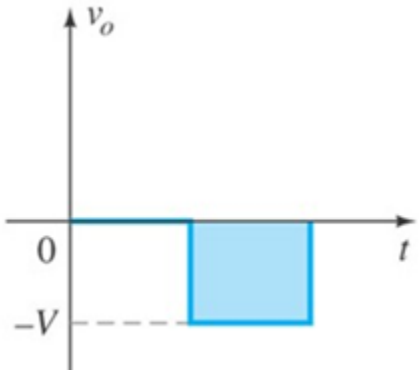


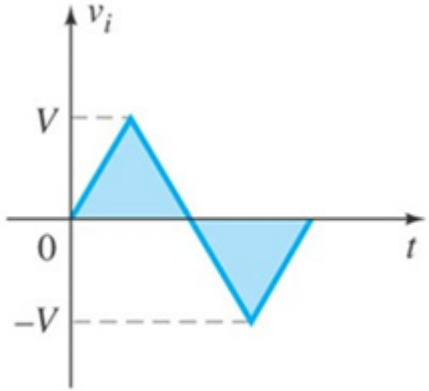


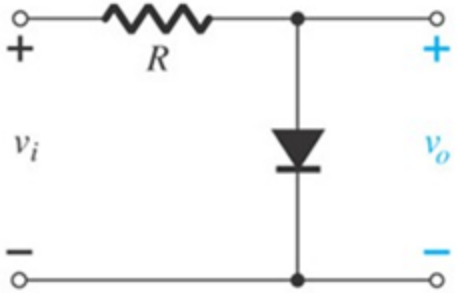
Parallel Clippers

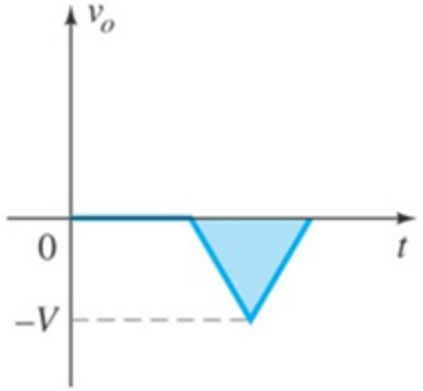








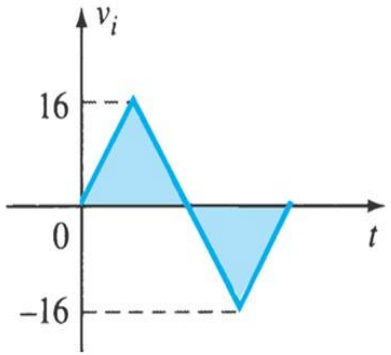


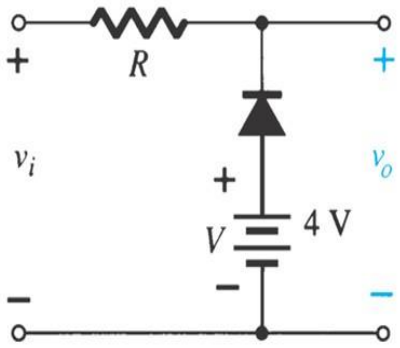


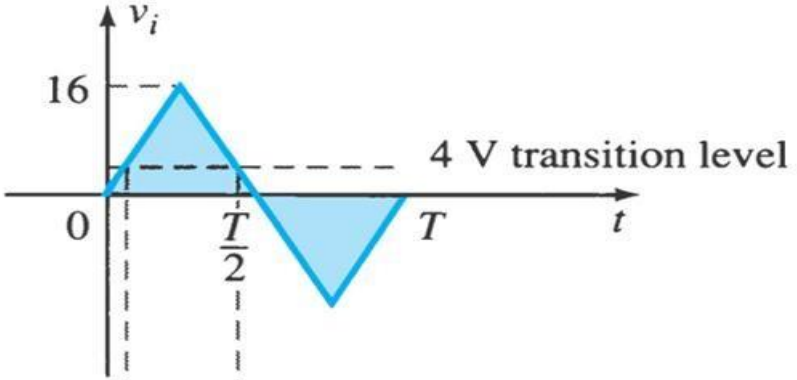
Example

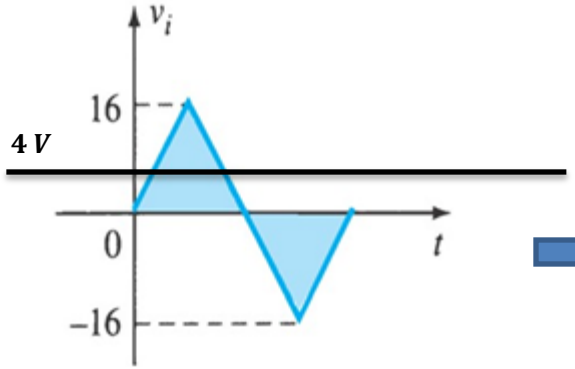
Determine v_o for the network shown (**Assume ideal diode**).

Solution

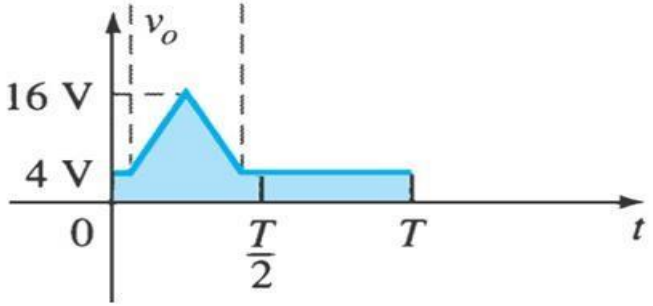








We have 2 regions to check:

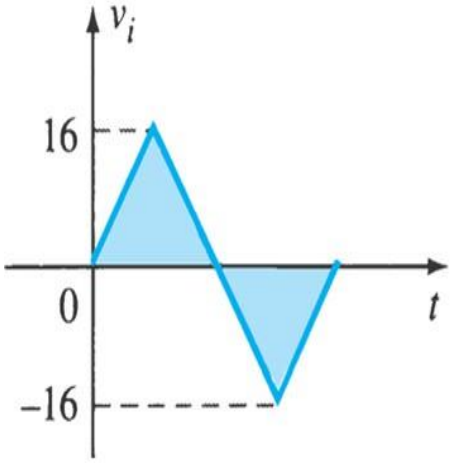


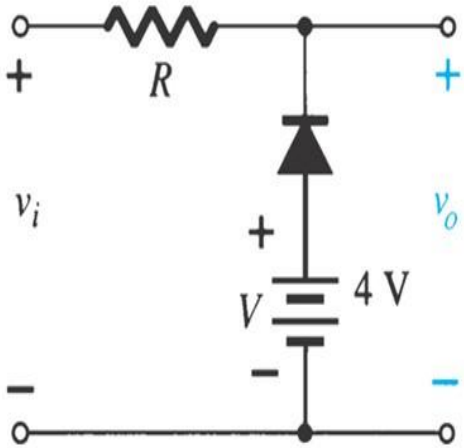
- $At\ v_i < 4 \rightarrow diode\ \boldsymbol{ON} \rightarrow v_o = 4\ V$
- $At\ 4 < v_i < 16 \rightarrow diode\ \boldsymbol{OFF} \rightarrow v_o = v_i$

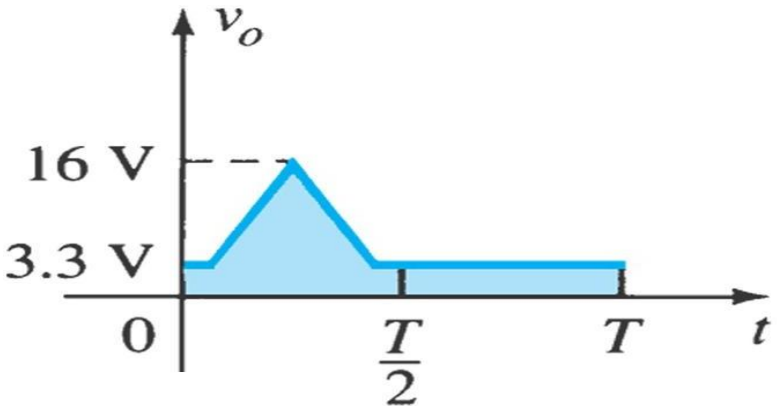
Example

Repeat the previous example assuming practical diode

Solution



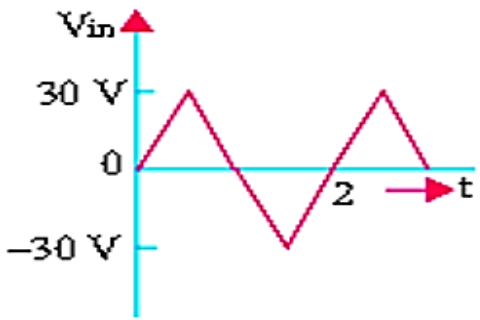


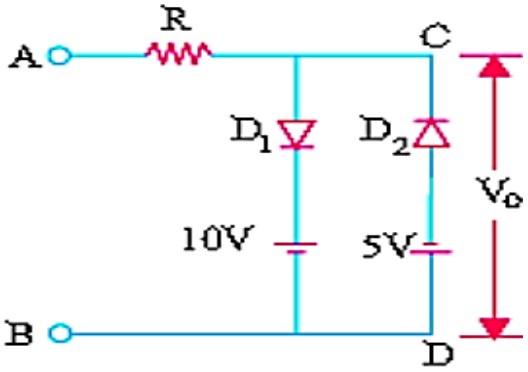


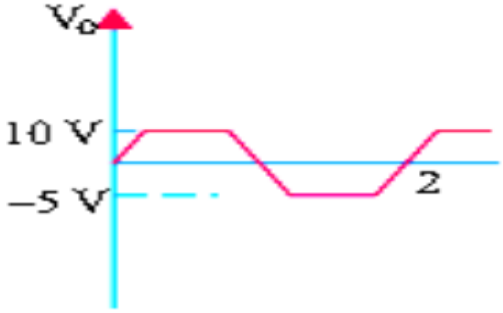
Example

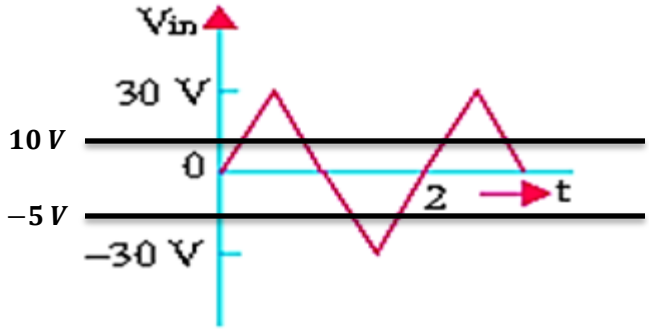
Determine v_o for the network shown (**Assume ideal diode**).

Solution





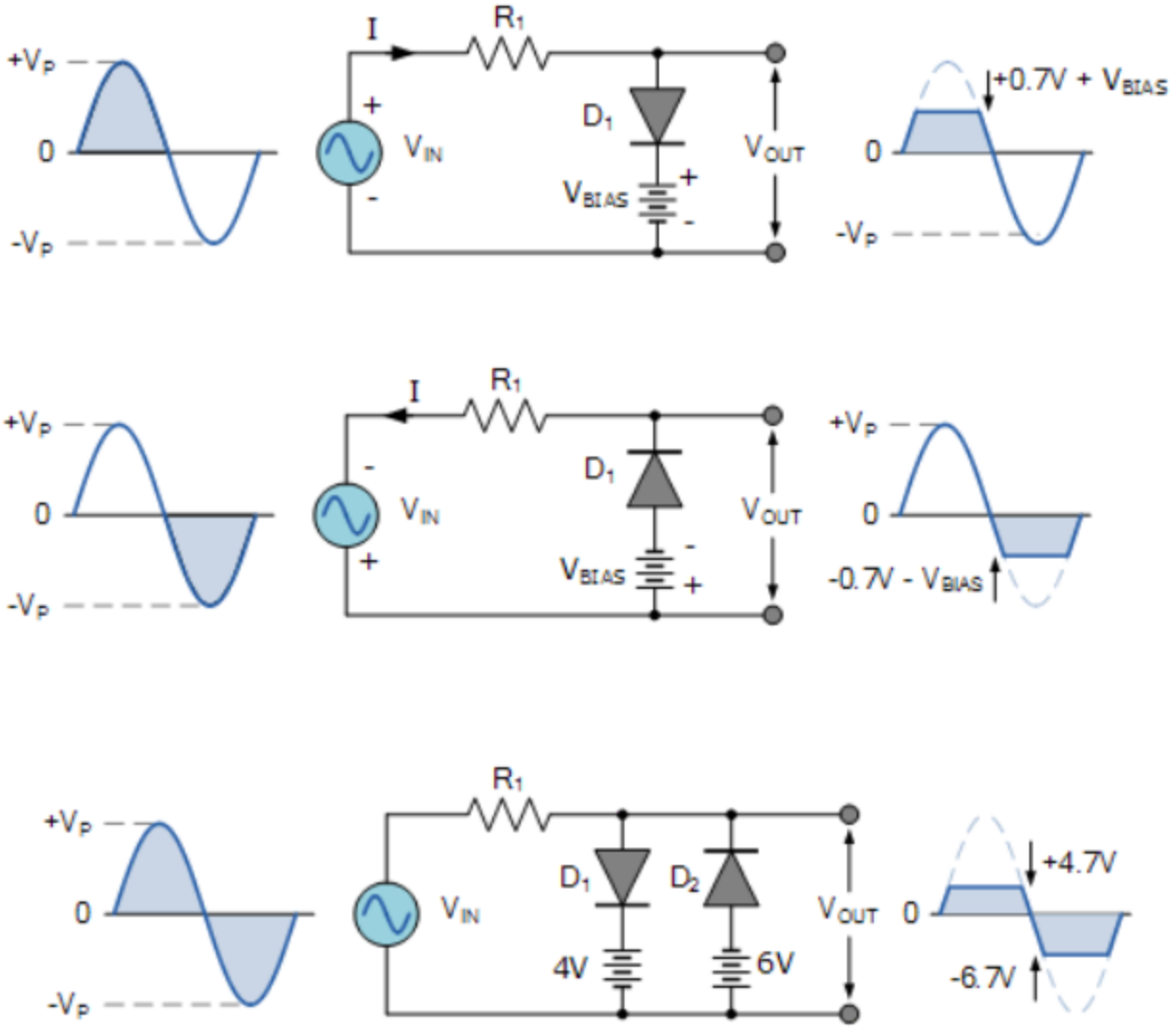




We have 3 regions to check:

- $-5 < v_i < 10 \rightarrow D1$ ***OFF and*** $D2$ ***OFF*** $\rightarrow v_o = v_i$
- $v_i > 10 \rightarrow D1$ ***ON and*** $D2$ ***OFF*** $\rightarrow v_o = 10\ V$
- $v_i < -5 \rightarrow D1$ ***OFF and*** $D2$ ***ON*** $\rightarrow v_o = -5\ V$

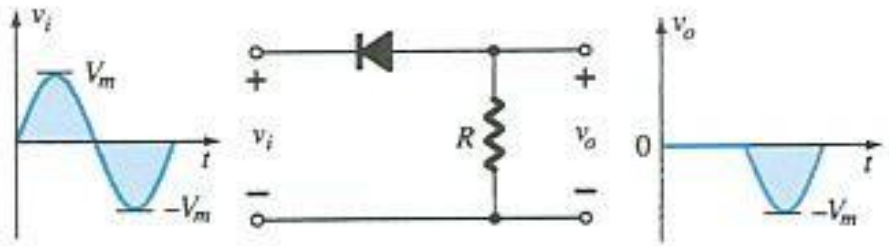
Examples (assume practical diodes)



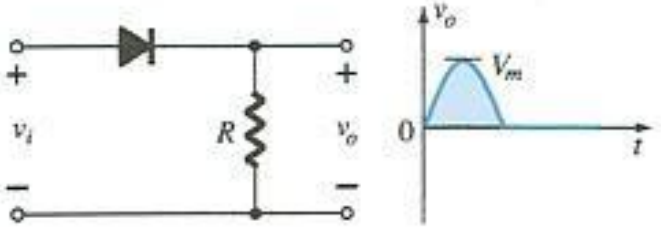
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

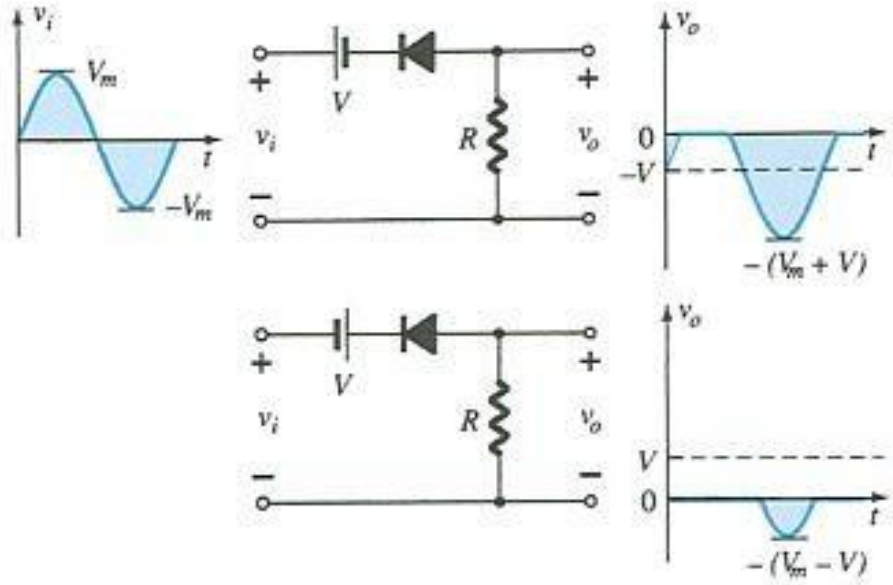
POSITIVE

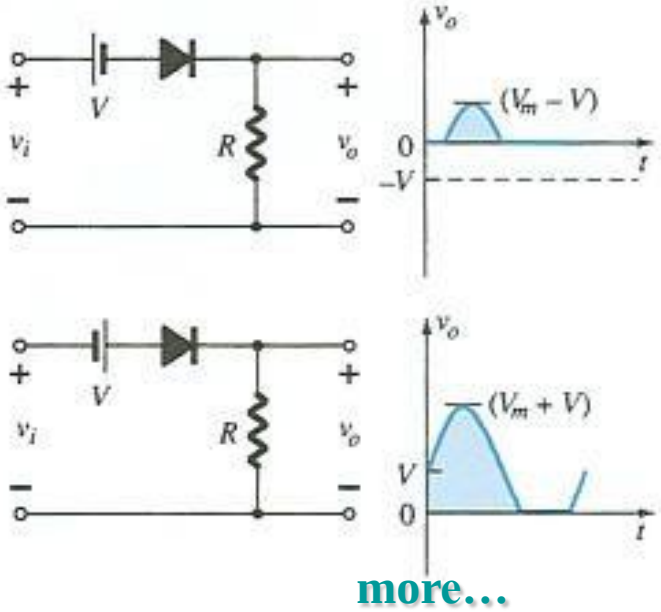


NEGATIVE



Biased Series Clippers (Ideal Diodes)

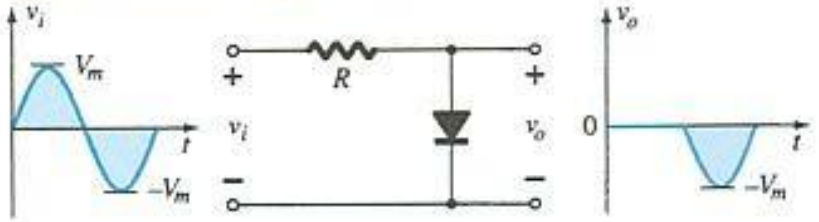


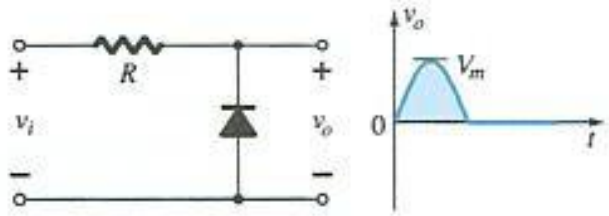


more…

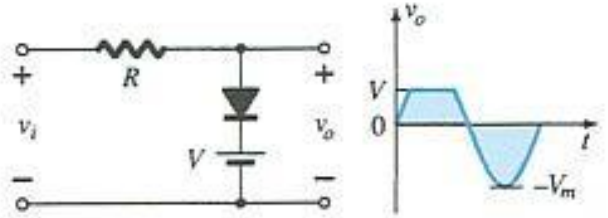
Summary of Clipper Circuits

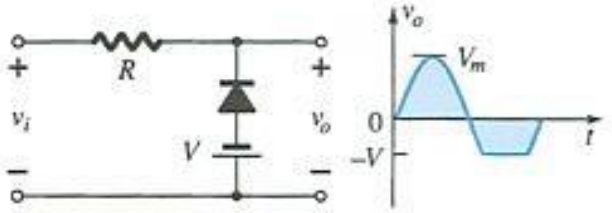
Simple Parallel Clippers (Ideal Diodes)

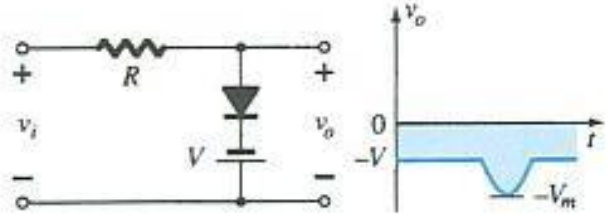


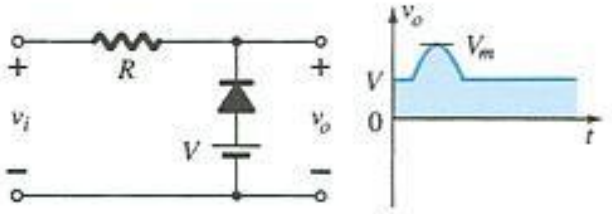


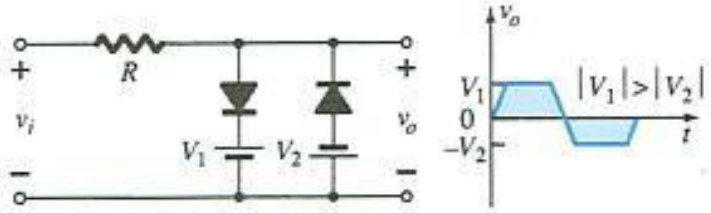
Biased Parallel Clippers (Ideal Diodes)











Task for students

The previous two slides present additional structures of **clipping circuits**. Your objective is to conduct a **comprehensive analysis** of these circuits by following these steps:

Circuit Analysis:

- Assume appropriate component values (**resistors, diodes, voltage sources, etc.**).
- Perform a theoretical analysis to determine the expected output waveform.

Verification and Simulation:

- Use Proteus**,** to model the circuits.
- Apply a suitable input signal (**sinusoidal or triangular**) and observe the output.
- Compare the simulated results with your theoretical predictions.

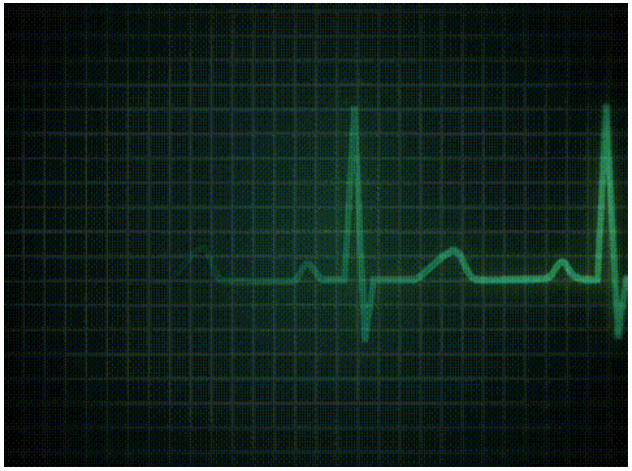
(3) Clampers

Real time applications of clampers

1. Restores DC level in AM/FM receivers.
2. Stabilizes radar signals.
3. Adjusts TV signal baselines.
4. Corrects baseline drift in Electrocardiogram (ECG) machines.
5. Supports Electroencephalogram (EEG) and bio-sensor signal processing.
6. Calibrates oscilloscopes.
7. Protects circuits from negative voltage spikes.
8. Prevents audio distortion in amplifiers.
9. Aligns video signals for proper display.
10. Triggers pulse detection circuits.

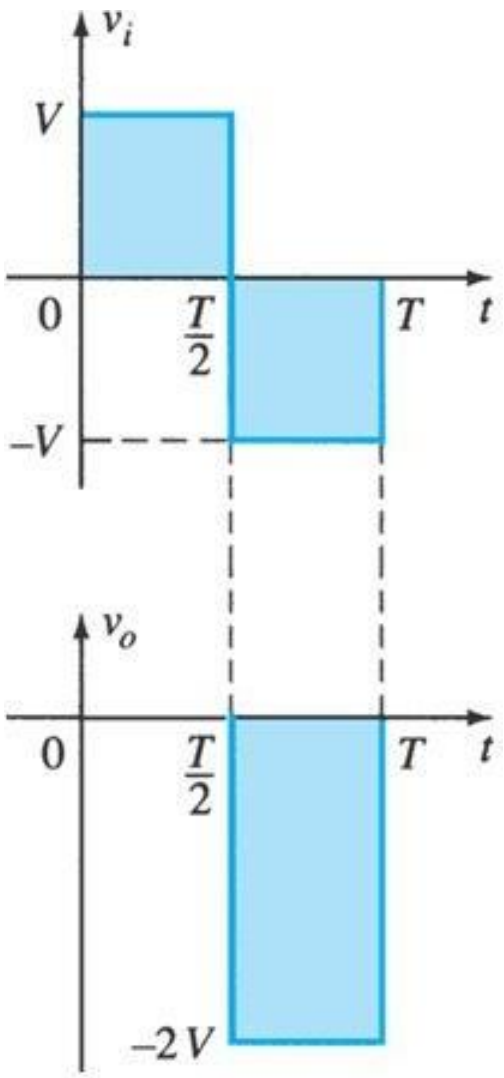


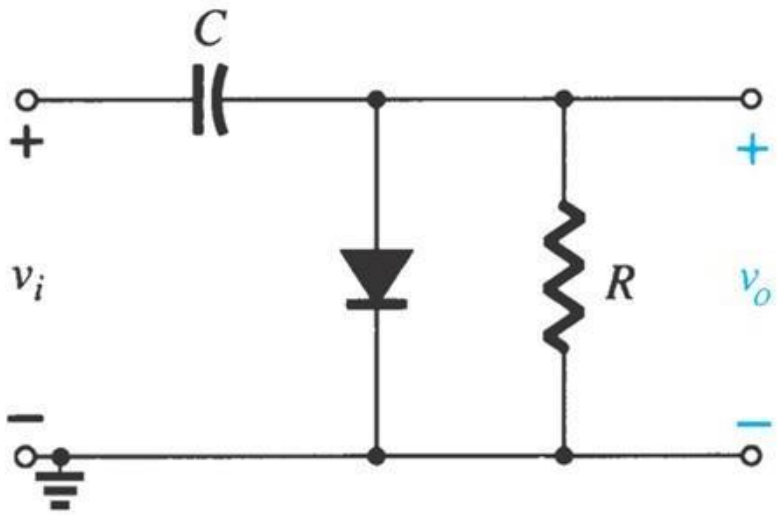
EEG



ECG

Clampers



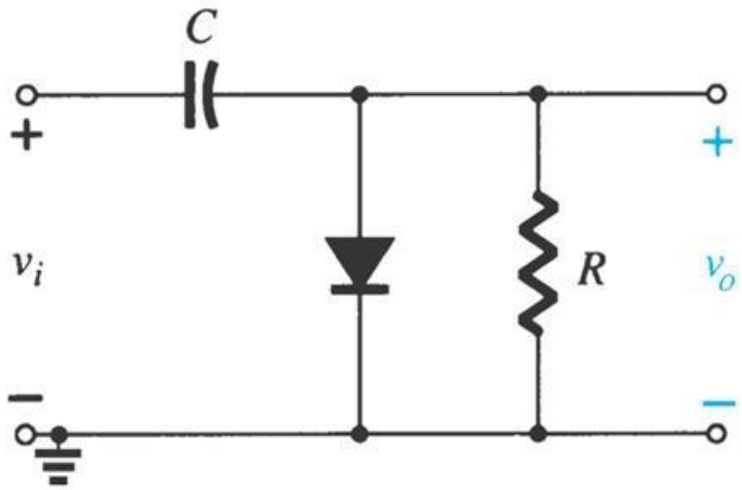


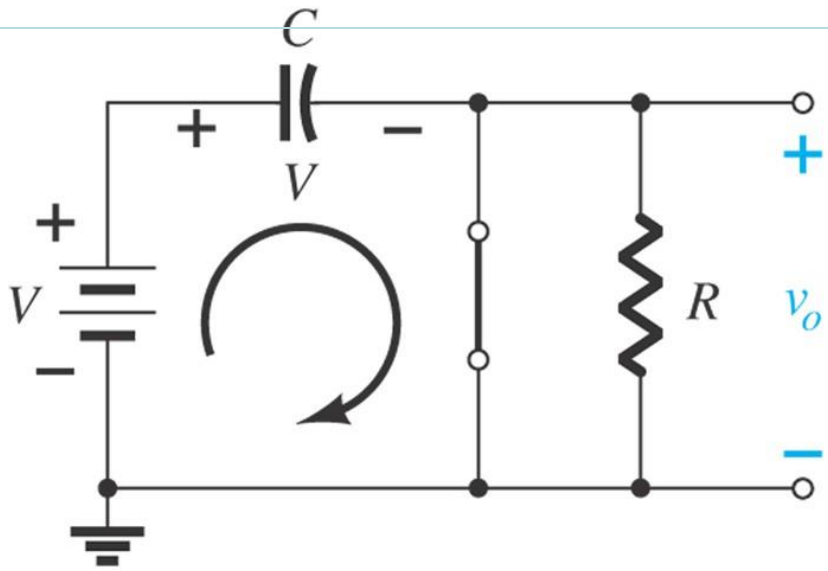
A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different DC level.

Clampers

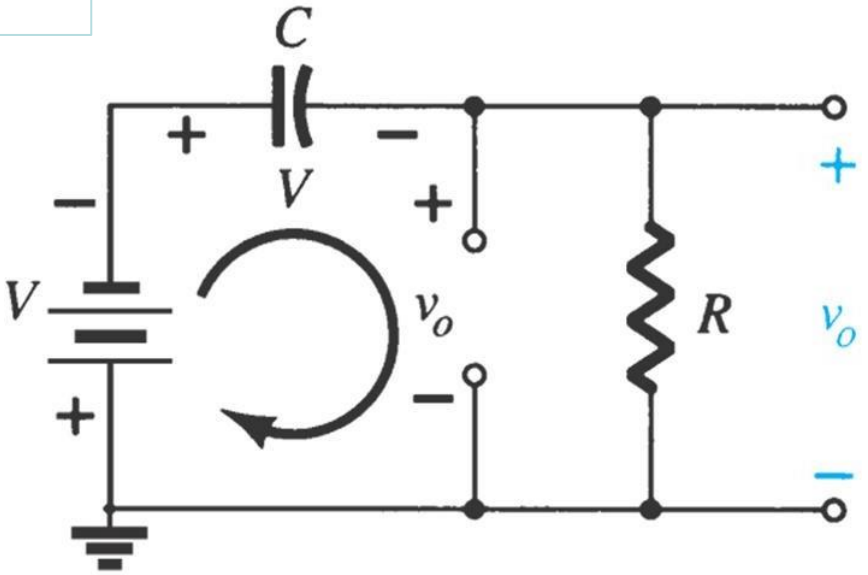
Conditions of analysis:

1. Start with AC input signal part (+ve or –ve part) which makes the diode ON
2. $\tau = RC \gg T\ (period\ of\ input\ signal) \rightarrow \boldsymbol{\tau \gg 10T}$



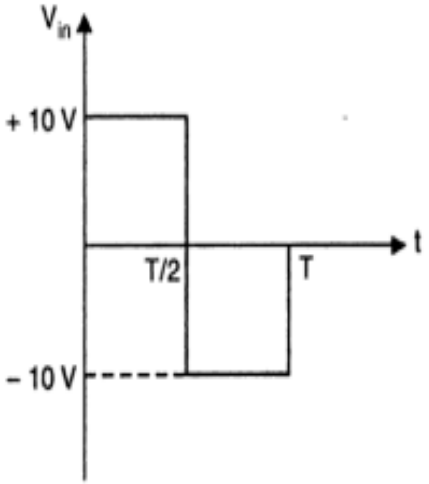


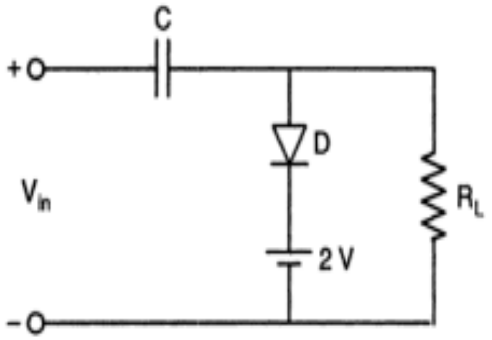
Diode "on" and the capacitor charging to V volts.



Determining v_o with the diode "off."

What is the output voltage that you would expect to observe across R_L in the clamping circuit of the shown figutre. Assume that RC is large enough to prevent significant capacitor discharge.





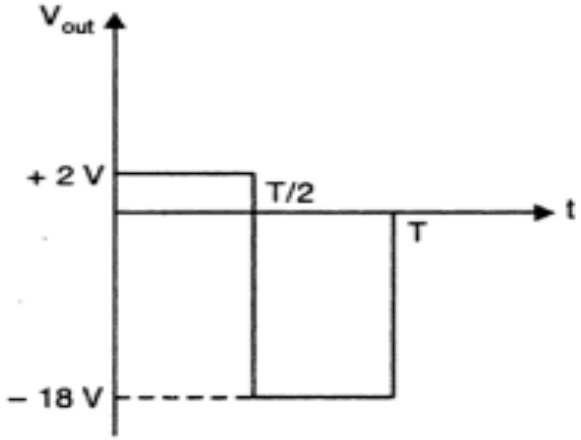
Assume ideal diode

Check conditions (at 10 V diode is on, condition is satisfied)

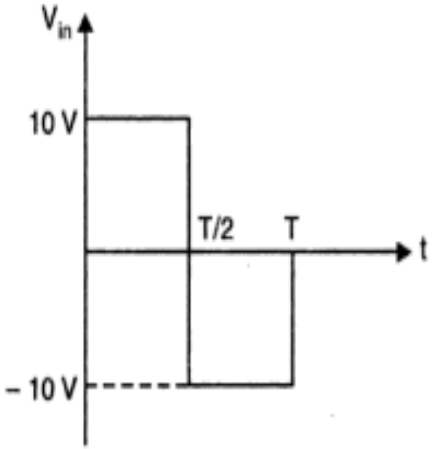
$$\mathbf{10 - Vc - 2 = 0}$$

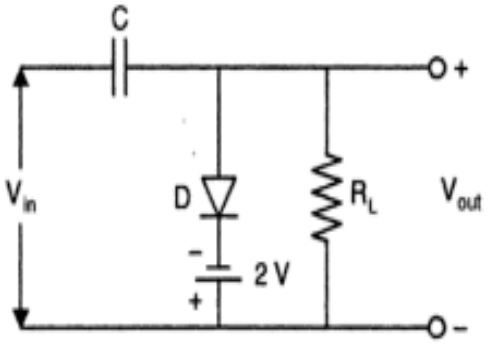
$$\boldsymbol{Vc = 8\,V}$$

Then shift **down** vertically with 8 V to get output



What is the output voltage that you would expect to observe across R_L in the clamping circuit of the shown figutre. Assume that RC is large enough to prevent significant capacitor discharge.



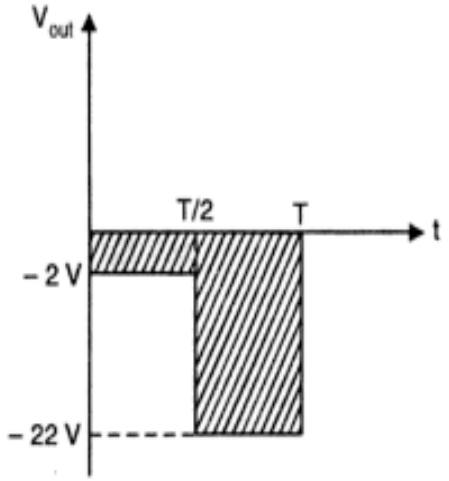


Assume ideal diode

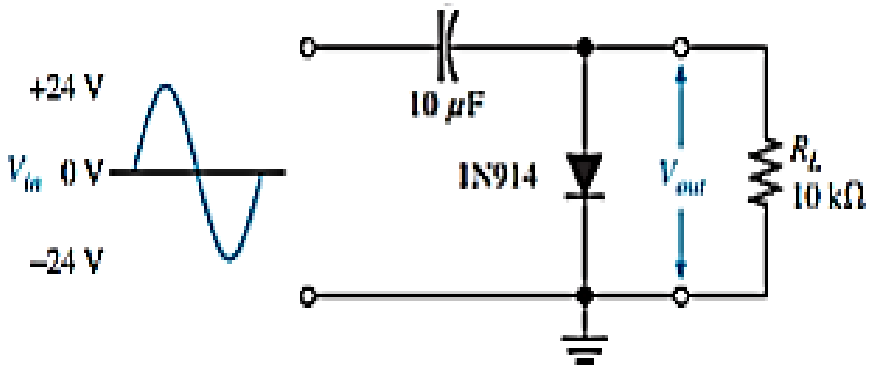
Check conditions (at 10 V diode is on, condition is satisfied)

$$\mathbf{10 - Vc + 2 = 0}$$
$$\boldsymbol{Vc = 12\ V}$$

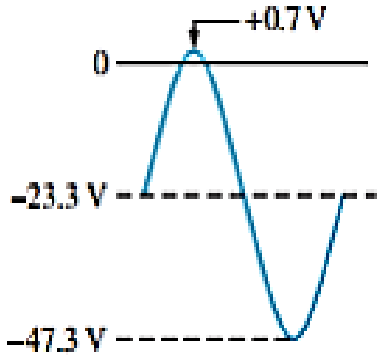
Then shift **down** vertically with 12 V to get output



What is the output voltage that you would expect to observe across R_L in the clamping circuit of the shown figutre. Assume that RC is large enough to prevent significant capacitor discharge.



Solution:



Assume practical diode

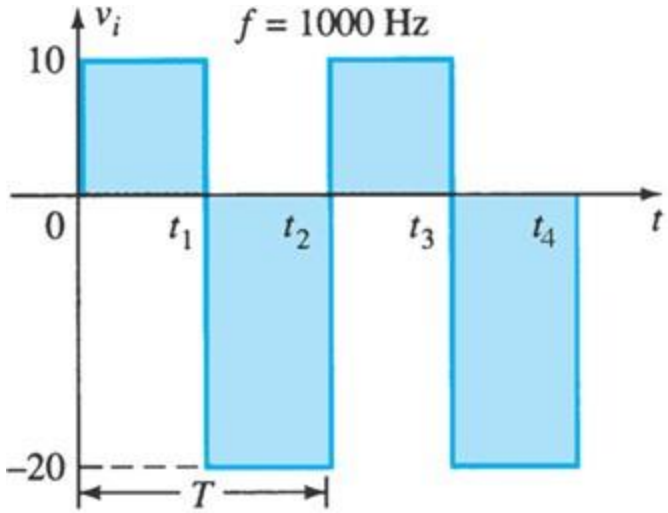
Check conditions (at 24 V diode is on, condition is satisfied)

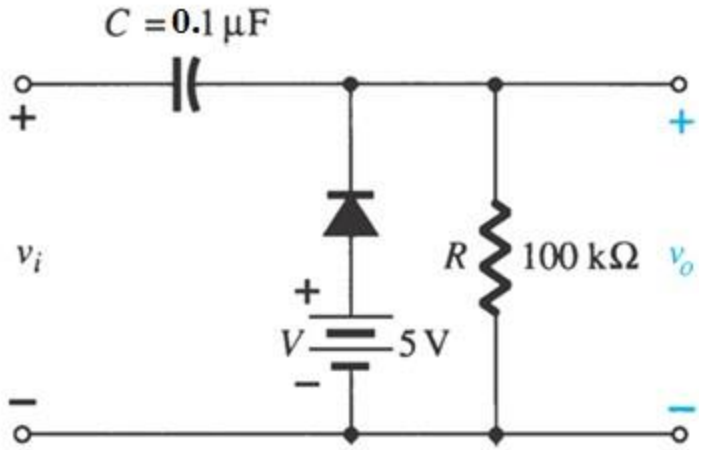
$$24 - Vc - 0.7 = 0$$
$$Vc = 23.3\ V$$

Then shift **down** vertically with 23.3 V to get output

Example

Determine v_o for the network.



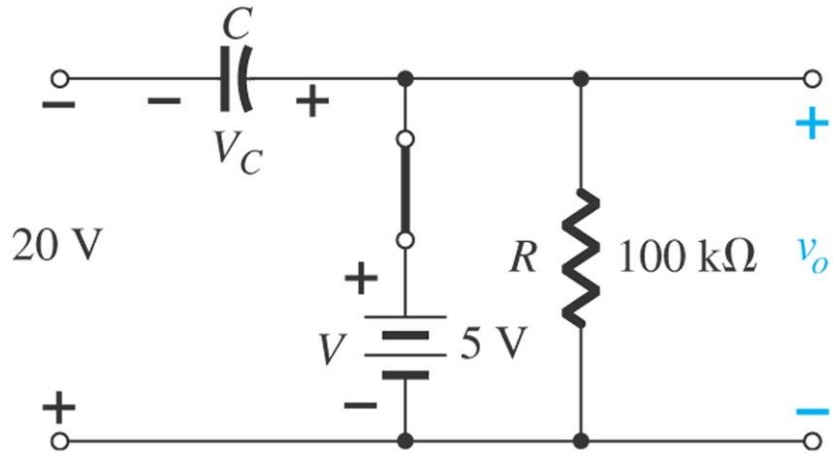


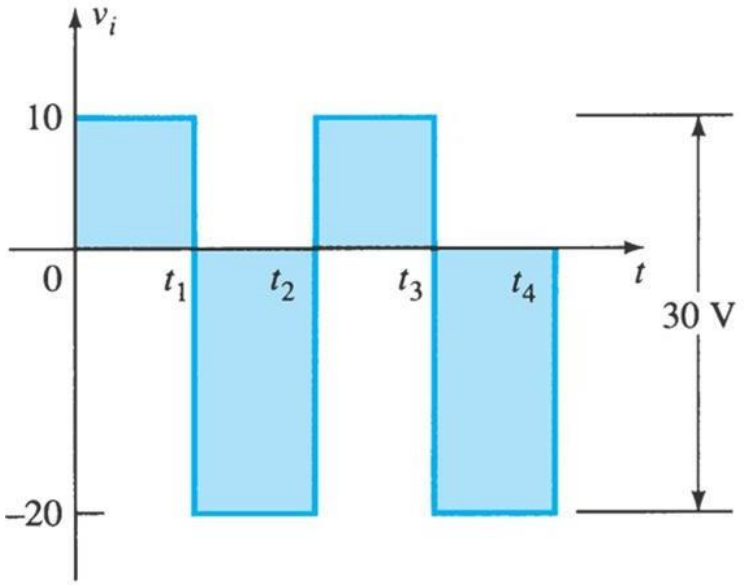
Solution

Check conditions:

at (10 V), diode is **off** (condition **is not satisfied**)

Example



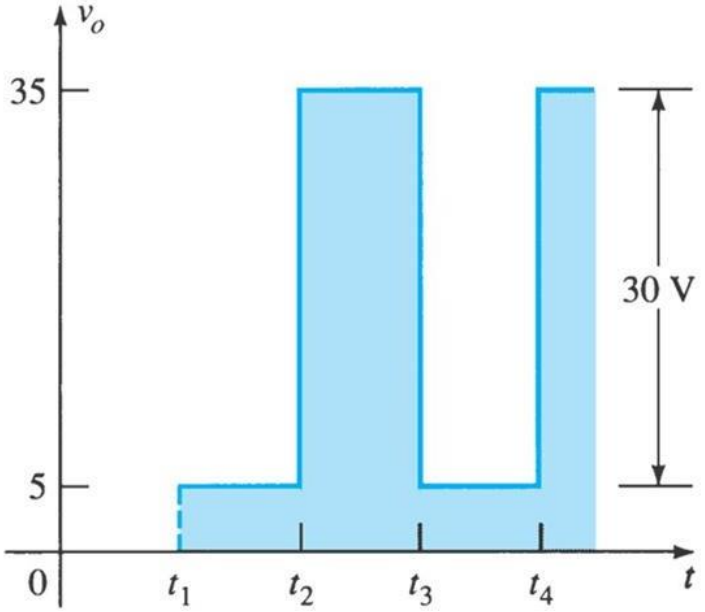


- At $(-20\ \text{V})$, diode is on,

$$20 - Vc + 5 = 0$$

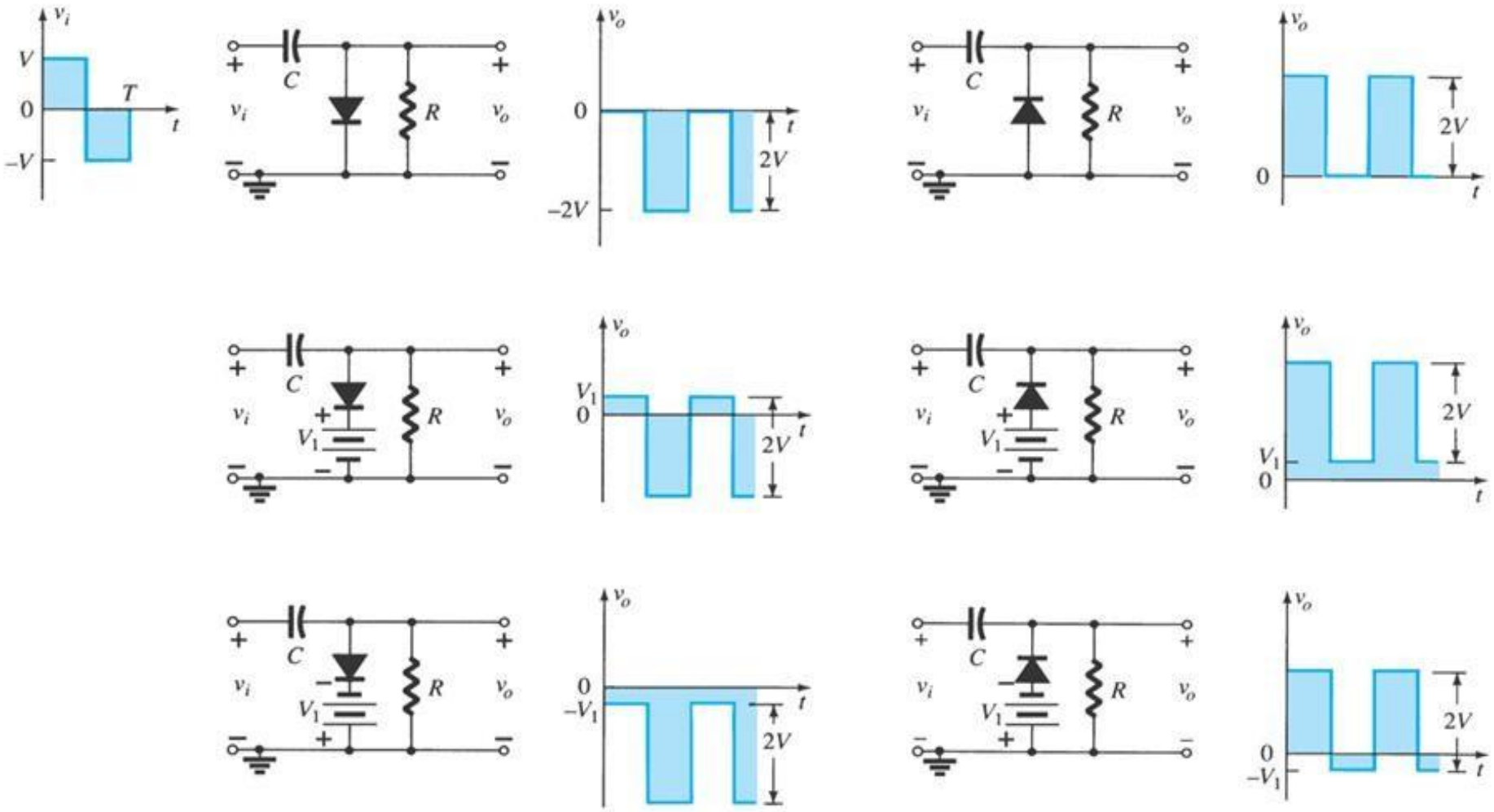
$$\text{Vc} = 25\ \text{V}$$

Then shift **up** vertically with 25 V to get output (except first cycle as diode was off and condition is not satisfied there)



Summary of Clamper Circuits

Clamping Networks



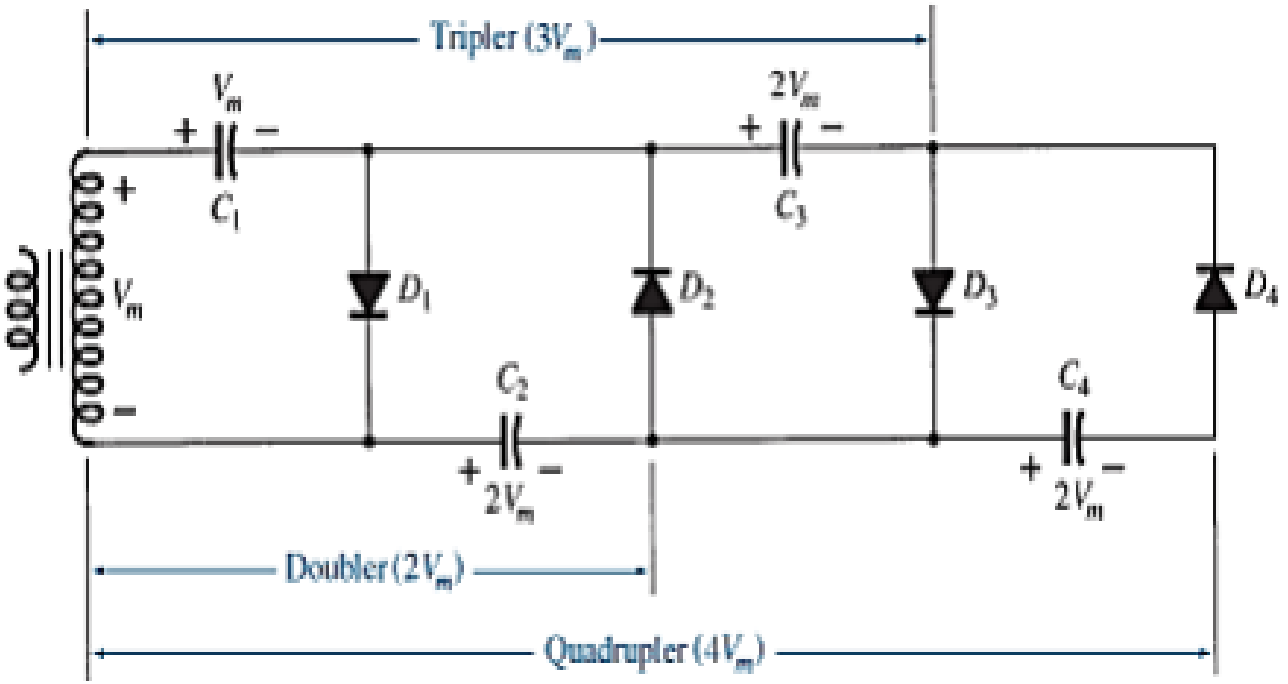
Voltage Multipliers

1. **X-ray machines** – Generates high voltage for medical imaging.
2. **CRT displays** – Powers old TVs and oscilloscopes.
3. **Particle accelerators** – Provides high DC voltage for scientific research.
4. **Electric fences** – Steps up voltage for effective operation. In **electric fences**, voltage multipliers are used to **boost** a low input voltage (e.g., from a battery or mains power) to a **much higher voltage** typically **5,000 to 10,000 volts** but with **very low current**. The high voltage creates a **short, harmless shock** that **deters animals or intruders** without causing injury
5. **Photomultiplier tubes** – Enhances light detection in sensitive equipment.

D. VOLTAGE MULTIPLIERS

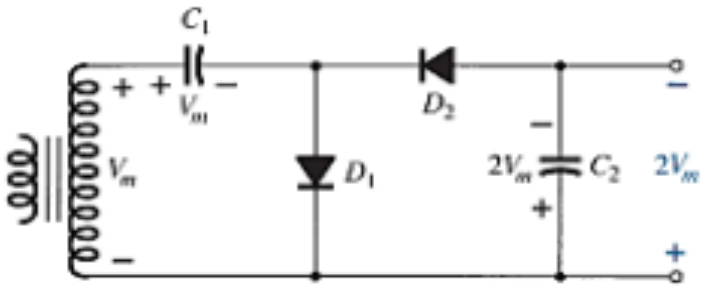
Voltage-multiplier circuits are employed to maintain a relatively low transformer peak voltage while stepping up the peak output voltage to two, three, four, or more times the peak rectified voltage.

The shown figure represents the main three types of voltage multipliers in one schematic diagram. We will discuss an example for this application (voltage doupler)



Voltage doupler (assume ideal diode model)

Half-wave voltage doubler



During the (+ve) half cycle:
Secondary diode D_1 conducts and diode D_2 is cut off charging capacitor C_1 up to the peak rectified voltage (V_m). Diode D_1 is ideally a short during this half-cycle, and the input voltage charges capacitor C_1 to V_m

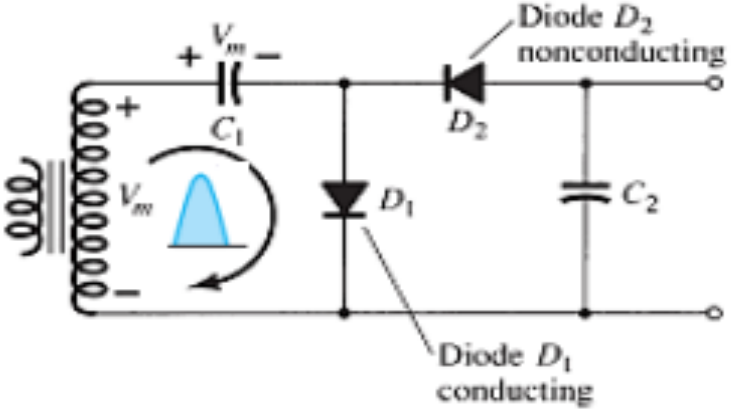
During the (−ve) half cycle:
diode D_1 is cut off and diode D_2 conducts charging capacitor C_2. Since diode D2 acts as a short during the negative half-cycle (and diode D_1 is open), we can sum the voltages around the outside loop

$-V_m - V_{C1} + V_{C2} = 0$

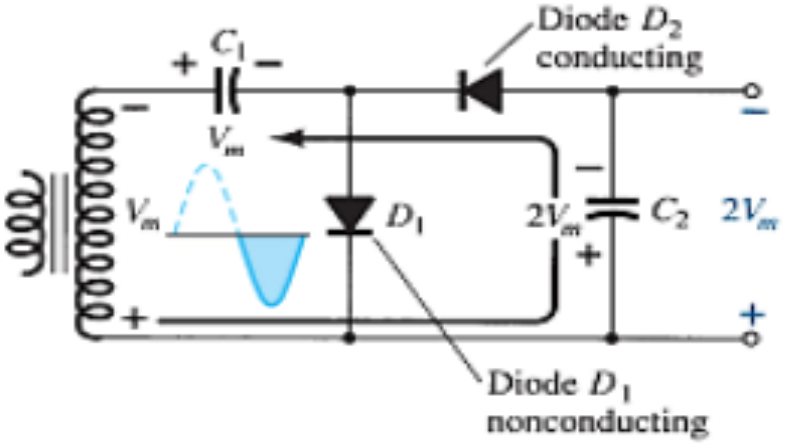
$-V_m - V_m + V_{C2} = 0$

Accordingly, $V_{C2} = 2V_m$

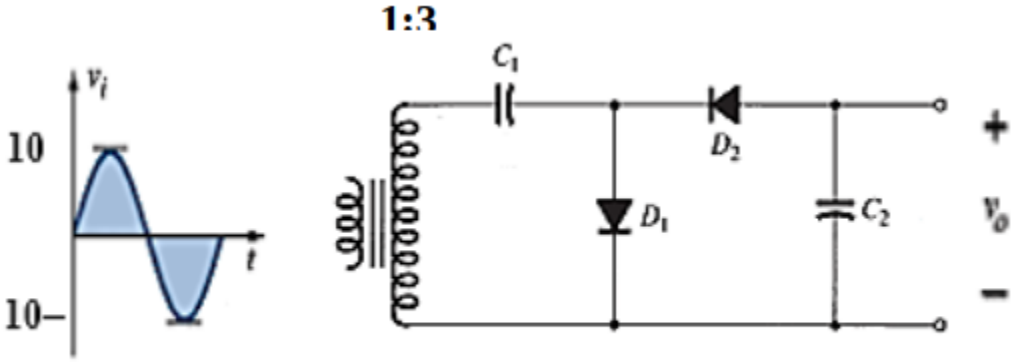
(+ve half cycle)



(−ve half cycle)



Example: Find the output



- This is a (2 stages) multiplier
- Secondary voltage =30 V due to the given ratio (1:3)
- Final output=$2 \times 30 = 60V$

Note: if we add another stage (additional diode with additional capacitor), secondary voltage is multiplied by 3, and so on